

Abrar Ul Riyaz

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SUMMARY

- AI Engineer and Backend Developer with experience building **LLM-powered applications, agentic workflows, and scalable backend systems** using Python, Django, and modern AI frameworks.
- Skilled in developing **RAG pipelines, autonomous agents, tool-calling systems, and conversational AI**, with strong foundations in system design, problem solving, and backend engineering.
- Hands-on experience building and deploying **production-ready AI systems**, integrating LLMs, APIs, databases, and cloud workflows to create reliable end-to-end AI applications.

EXPERIENCE

DarwixAI

Apr. 2026 – Present

AI Intern

Remote

- Working on **AI voice agents, agentic workflows, and LLM-powered automation systems**.
- Developing backend services and APIs for **real-time AI communication pipelines**.
- Building and integrating **LLM agents, tool-calling workflows, and conversational AI systems**.
- Contributing to scalable AI products using **Python, Django, PostgreSQL, and cloud-based services**.

GAASH Research Lab National Institute of Technology (NIT) Srinagar

Apr. 2025 – April. 2026

Research & Backend Development

Srinagar, India (Onsite)

- Conducting applied research in **Machine Learning, LLM reasoning, and Agentic AI systems**, focusing on building autonomous and tool-using AI workflows.
- Training, fine-tuning, and evaluating deep learning models on **NVIDIA H100 GPUs**, optimizing performance and scalability.
- Designing and developing **agentic pipelines (RAG, tool-calling, multi-step reasoning)** using modern LLM frameworks.
- Leading backend development for AI-driven applications, handling **database design, REST APIs, and scalable system architecture**.
- Built and deployed **LLM-powered full-stack systems** using **Django, PostgreSQL, and vector databases**, ensuring reliability and maintainability.

AI4BHARAT Lab Indian Institute of Technology (IIT) Madras

Dec. 2024 – Mar. 2025

Research Intern

Remote

- Conducted **Optical Character Recognition (OCR)** research on 3 Indian Languages: **Urdu, Bengali, and Assamese**.
- Worked with multiple OCR engines, including **AI4Bharats Surya OCR** and **Google Cloud Vision OCR**, for multilingual text extraction.
- Performed a detailed **comparative evaluation** of OCR engines based on accuracy, robustness, and language-specific performance.
- Documented experimental results and contributed to **dataset preparation, benchmarking reports, and performance analysis**.

TECHNICAL SKILLS

Programming Languages : Python, C, C++, JavaScript, TypeScript

Backend & APIs : Django, DRF, FastAPI, RESTful APIs, Pydantic, LangChain, LangGraph, LangSmith, MCP, RAG

Databases : PostgreSQL, MySQL, MongoDB, ChromaDB, Pinecone

Frontend : HTML, CSS, React, Tailwind CSS, Bootstrap, Streamlit

ML & Data Libraries : NumPy, Pandas, Scikit-learn, Matplotlib, OpenCV

Core CS : Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks

Tools & Platforms : Git, GitHub, Hugging Face, Postman, REST Client, Thunder Client

Deployment : Netlify, Vercel, Render

PROJECTS

Agenix [GitHub](#)

- * **Tech Stack**: Python, LangChain, Ollama, LLaMA 3.2, CLI
- * Developed an **agentic command-line interface** that translates natural language instructions into executable system commands.
- * Integrated **local LLM (LLaMA 3.2 via Ollama)** for offline inference, eliminating dependency on external APIs and improving privacy.
- * Built a **LangChain-based agent pipeline** to parse user intent and map it to structured shell commands (e.g., cd, ls, file operations).
- * Enabled **autonomous task execution** such as navigation, file management, and command chaining through natural language interaction.

PromptIQ [GitHub](#)

- * **Tech Stack**: Django, REST APIs, React, PostgreSQL, LLM APIs (GPT-4, Gemini)
- * Designed and developed a **full-stack AI chatbot** using Django and React, enabling dynamic selection and interaction with multiple **large language models**.
- * Implemented **PDF ingestion and document-based question answering**, allowing users to perform natural language queries and generate grounded AI-driven responses.
- * Built **context-aware conversational memory** to support multi-turn interactions and maintain coherent, stateful chat history across user sessions.

BIS-Basha [GitHub](#)

- * **Tech Stack**: Django, React.js, Sarvam API (Translation), GPT (Summarization), Google Cloud Text-to-Speech
- * Developed an **AI-powered multilingual web application** supporting **translation and text-to-speech (TTS)** for **22 Indian languages**, including Hindi, Tamil, and Bengali.
- * Implemented **document translation with structure preservation**, retaining original layout elements such as **tables, figures, headings, and formatting** after translation from English to Indian languages.
- * Integrated **Sarvam APIs** for high-quality **machine translation** and **Google Cloud Text-to-Speech** for natural-sounding speech synthesis across multiple languages.
- * Implemented **AI-driven PDF summarization** using **GPT-based models**, enabling fast and accurate understanding of uploaded documents.
- * Designed a **modular and scalable backend architecture** in Django to orchestrate translation, summarization, and TTS processing pipelines.

RagFirst [GitHub](#)

- * **Tech Stack**: React (TypeScript), Django, LangChain, Pydantic, OpenAI GPT-4, Vector Embeddings
- * Developed a **retrieval-augmented generation (RAG)** application for **document and video understanding**, enabling grounded and context-aware AI responses.
- * Implemented **multi-document PDF ingestion** with **semantic search** over vector embeddings, providing accurate AI-generated summaries and information retrieval.
- * Built a **context-aware conversational chatbot** supporting **multi-turn question answering** using conversation history and retrieval context.
- * Integrated **YouTube video analysis** by generating concise summaries and enabling **natural language Q&A** over video transcripts.
- * Designed an **end-to-end retrieval and reasoning pipeline** using LangChain and Pydantic to ensure structured inputs, grounded outputs, and reliable response generation across heterogeneous data sources.

PUBLICATIONS

Grounded Surgical Benchmark for Language-Conditioned Tool Segmentation [GitHub](#) [arXiv](#)

- * **Conference:** MICCAI 2026 (Under Review)
- * Built a **grounded VLM benchmark (600+ images, 1000+ annotations)** for tool localization from language.
- * Evaluated **VLMs** (Qwen-VL, Gemma, InternVL) on segmentation and grounding.

QTrack: Query-Driven Reasoning for Multi-modal Multi-Object Tracking [GitHub](#) [arXiv](#)

- * **Conference:** ECCV 2026 (Under Review)
- * Framed tracking as **language-driven spatiotemporal reasoning**.
- * Proposed **QTrack (RL + TAPO)** and **RMOT26 benchmark**.
- * Achieved **SOTA** in spatial accuracy and temporal consistency.

SurgReason: Reasoning-Driven Language-Conditioned Surgical Tool Segmentation [GitHub](#) [Paper](#)

- * **Conference:** ECCV 2026 (Under Review)
- * Proposed a **reasoning-based segmentation pipeline** (which, why, where).
- * Used **SFT + RL** for alignment and robustness.
- * Improved **BBox IoU** and mask quality (**reasoning > scaling**).

EDUCATION

National Institute of Technology (NIT) Srinagar

Nov. 2022 – Jul. 2026

Bachelor of Technology in Information Technology

Srinagar, India

- * **CGPA:** 8.27 / 10
- * **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, Machine Learning

Nigeen Higher Secondary School

Dec. 2019 – Mar. 2021

Higher Secondary (HSC / 10+2)

Srinagar, India

- * **Percentage:** 94.6%
- * **Subjects:** Mathematics, Physics, Chemistry

LANGUAGES

Languages : English (Professional Proficiency), Urdu (Native), Hindi (Native)